

Safe lifting techniques

Table of contents

1. BYCN Directive and Life-Saving-Rules.....	3
2. Operational requirements	3
2.1. Requirements before the lift	3
2.2. Requirements during the lift	3
2.3. Suspended & travelling loads	3
2.4. Lifting operator with no continuous view on the load	4
2.5. Blind lift.....	4
2.6. Guiding of the load	4
2.7. Multiple work equipment lift (Tandem lift)	5
2.8. Prohibited maneuvers	5
2.9. 3-times rule.....	5
3. Slinging practices	5
3.1. Choice of lifting accessory	5
3.2. Choice of slinging methodology	6
3.3. Wire rope sling assembly.....	8
3.4. Evaluate the centre of gravity (COG).....	8
3.5. Attaching the load	8
3.6. Areas of vigilance.....	9
4. Management of lifting accessories.....	10
4.1. Lifting accessories register	10
4.2. Marking.....	10
4.3. Equipment and accessories examination	10
4.4. Periodic general inspection	11
4.5. Visual inspection of equipment and accessories.....	12
4.6. Wire rope slings.....	12
4.7. Chain slings	12
4.8. Webbing slings.....	13
4.9. Storage of lifting accessories	14

5. Specific lifting equipment and accessories	15
5.1. Metal sheet handling clamps	15
5.2. Vacuum spreader beam and vacuum pump	15
5.3. Remote controlled hooks	16
5.4. Lifting magnets and electromagnets	16
5.5. Segment lifters, barrel lifters, jib cranes	16
5.6. Eye bolt	17
5.7. Articulated hoist rings	17
5.8. Construction plants lifting platforms	18
5.9. Big bags	18
5.10. Storage baskets	19

1. BYCN Directive and Life-Saving-Rules

The Safe Lifting Techniques depicted in this document came from :

- The BYCN Lifting Directive
- The Life Saving Rules:
 - I ensure that the load being lifted is a **single unit**
 - I do not stand between a load and a **fixed obstacle**, or **under the load** when it is being **lifted up or lowered down**
 - I use lifting **accessories** that are **checked** and **suitable** for the load
 - I do not enter an **exclusion zone**

2. Operational requirements

2.1. Requirements before the lift

Before authorizing any lifting, the Signaller ensures that :

- the conditions foreseen in both the *Lifting Equipment Utilization Plan* and the *Lifting Operation Plan* are in place and the type of lifting has been properly evaluated
- loads are correctly attached or bound together with restraint devices together into one piece, and appropriately slung / rigged, otherwise they are placed in a 5-sided container or basket preventing any elements from falling off during the lifting operation
- the load is correctly attached/slung, in accordance with the *Lifting Operation Plan*
- weather conditions are compatible with the provisions set out in the *Lifting Operation Plan*
- the environment of the maneuver is sufficiently clear that there is no risk of colliding with nearby equipment/plant/structures/utilities
- the Slinger(s) has(have) safe access to lifting points to attach and detach the load
- lifting equipment is set up, level and supported as per Lifting Operation Plan (outriggers, mats, etc.)
- load receptions areas are clear

2.2. Requirements during the lift

- only the Signaller has the authority to allow a load to be slung or unslung, and to give instructions to the lifting operator
- the Signaller ensures no one is under the load when raising and lowering a load, as well as between the load and a fixed object
- the Signaller authorizes the slings to be slackened and then for the load to be unslung after ensuring the load is stable

2.3. Suspended & travelling loads

- it is forbidden to carry loads over people when raising or lowering the load
- a safety perimeter is set up to minimize the consequences of falling objects during the lift. All ongoing work within this safety perimeter will be stopped during all stages of the lift
- where the load is lifted or moved near any public accessway, road or rail or other route of travel, special measures are taken to guard against any danger that may arise should the transported load fall

- no load shall travel “over” aforementioned locations without them having been confirmed to have been closed off to access
- for tower cranes, prohibited zones must be set prior to their putting into service via the anti-collision management system

2.4. Lifting operator with no continuous view on the load

The lifting equipment control station/cabin is set up so that the Operator can clearly see the maneuvers made by mobile parts of the lifting equipment. If this is not possible at all stages, then :

- a Signaller provides continual direction to the Operator if he cannot see the load or if any part of the lift is not visible to him either directly or by means of assistance devices (cameras and others). The Signaller is assisted, if necessary, by one or more other Signallers stationed at locations where they can watch the movement of the suspended load
- organizational measures are taken to prevent collisions that could endanger people
- where practicable, use lifting equipment fit with cameras installed on hook blocks

2.5. Blind lift

- lifts to be executed when the Operator has no line of sight of the load to be lifted, or of the areas where the load is to be set down or picked up, are directed by the Lifting Supervisor who issues instructions to the duly appointed Signaller, who in turn is in direct radio contact with the Operator of the lifting equipment
- the Lifting Supervisor may appoint a second Signaller if the latter is unable to be in both the load pick-up and set-down locations. Both Signallers must maintain a radio link with the lifting equipment Operator. A communication protocol must be clearly defined to ensure the Operator always receives orders from only one designated Signaller
- the Signaller must not under any circumstances physically participate in the lifting operation: guiding, attaching, detaching the load, etc... He must only act as the eyes and communicator of signals to the lifting equipment Operator
- before starting any blind lift operation, a hold point is automatically observed to ensure that the personnel and material means are adequate and that instructions have been shared and understood by all personnel involved. Only the Lifting Supervisor is authorized to release the hold point

2.6. Guiding of the load

- the Slinger has safe access to the slinging and unslinging points of the load
- during slinging operations, no lifting equipment maneuver is allowed until the Slinger has given his go-ahead to the Signaller
- it is forbidden to swing a load towards a set down point that cannot normally be reached by the lifting equipment
- when guiding approaching loads, no one should touch the load and the guiding should be done from a distance using a tag line or a push-pull pole long enough for the Slinger to maintain a safe distance from the load
- devices (for example handles or clamps) should be made available to Slingers to help control the load and avoid risk of pinching

2.7. Multiple work equipment lift (Tandem lift)

A lift involving the simultaneous use of 2 tower cranes is dangerous and prohibited.

2.8. Prohibited maneuvers

- forbidden to disable or bypass/override any safety device or use lifting equipment where these items are not fully functioning
- no lifting of multiple items not tied or bound together
- no side-pull lifting or side pulling
- no lifting of loads that are not free of the ground or its support.
- no lifting of people, except in the case of evacuation using an accessory designed for this purpose (rescue evacuation cage/stretchers)
- no lifting of construction site machinery unless it is attached or rigged to the manufacturer or suppliers approved designated “lifting points” provided for such purpose or on an appropriate designed for purpose transport platform
- forbidden to leave a suspended load unattended
- No load, including lifting accessories, can be left suspended to a lifting equipment that is not in use

2.9. 3-times rule

1.
Attach the lifting accessories without lifting the load and let the slinger move away – minimum 3



3.
Lift the load to its transfer height

2.
Slowly apply tension and slightly lift the load (at least 300mm) to check the rigging position, the balance of the load and the resistance of accessories at least for 3 seconds

3. Slings practices

Slings include all operations involving attaching lifting accessories to a load and to lifting equipment. Lifting accessories arrangement is generally made up of one or more slings (wire rope, chain or webbing) but may also include rigid elements such as lifting/spreader beams, clamps, etc.

3.1. Choice of lifting accessory

Choosing the right sling or accessory is one of the trickiest aspects of slings preparation. This choice is crucial for the safety of the lifting operation.

The materials (steel, webbing) used will have an impact on the weight of the sling and its flexibility. The working environment, the location of use and the type/shape of load (protection required, anchoring point) may lead to different choices of material for the slings.

3.2. Choice of slinging methodology

The selection of the most appropriate slinging methodology for a given load to be lifted requires a specific analysis by the lifting Supervisor and/or the Method engineers. It should be noted that standard rigging arrangements are proposed in the appendix **BYTP-H&S-INF-2141 [Standard Slinging]** and that they can be used as they are, in the concerned *Lifting Operation Plan*. Any necessary modification of a standard slinging leads to define a specific slinging.

Any slinging arrangement not detailed in the above appendix shall be designed, as a minimum, according to the following criteria : The lifting capacity of a sling against a particular method of slinging is calculated by multiplying the WLL of the single sling by the slinging mode factor M.

Always check that the mass of the load is less than WLL x M

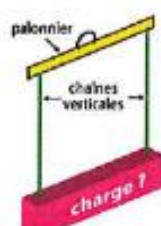
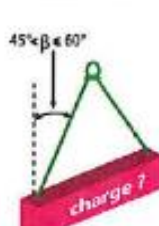
This mass is a function of the density (weight per unit volume) of the material of which it is made. Main densities to keep in mind are listed in the following table :

Material	Weight / m ³	Material	Weight / m ³
Lumber	600 Kg	Rubble	1200 Kg
Hollow bricks (15 cm thick)	900 Kg	Rubble stone (siliceous or calcareous)	1500 to 1700 Kg
Solid bricks	2000 Kg	Granite stone	17300 to 2300 Kg
Concrete blocks	2000 to 3300 Kg/unit	Earth, sand (dry)	1800 Kg
Concrete	2300 Kg	Steel	7850 Kg
Reinforced concrete	2500 Kg		

Elingues textiles		Charge Maximale d'Utilisation / CMU (kg) Elingue ronde, plate ou élingue à 1 brin / 2 brins										
		Angle β										
		Direct	Noeud coulant	En panier Type U Max 7°	En panier Type U 7° - 45°	En panier Type U 45° - 60°	En panier Type O 7° - 45°	En panier Type O 45° - 60°	Direct 45°	Noeud coulant 45°	Direct 45° - 60°	Noeud coulant 45° - 60°
	Elingue Ronde											
	Sangle plate											
	Facteur de mode	1,0	0,8	2,0	1,4	1,0	0,7	0,5	1,4	1,12	1,0	0,8
CMU	500 kg	500	400	1 000	700	500	350	250	700	560	500	400
	1 000 kg	1 000	800	2 000	1 400	1 000	700	500	1 400	1 120	1 000	800
	2 000 kg	2 000	1 600	4 000	2 800	2 000	1 400	1 000	2 800	2 240	2 000	1 600
	3 000 kg	3 000	2 400	6 000	4 200	3 000	2 100	1 500	4 200	3 360	3 000	2 400
	4 000 kg	4 000	3 200	8 000	5 600	4 000	2 800	2 000	5 600	4 480	4 000	3 200
	5 000 kg	5 000	4 000	10 000	7 000	5 000	3 500	2 500	7 000	5 600	5 000	4 000
	6 000 kg	6 000	4 800	12 000	8 400	6 000	4 200	3 000	8 400	6 720	6 000	4 800
	8 000 kg	8 000	6 400	16 000	11 200	8 000	5 600	4 000	11 200	8 960	8 000	6 400
	10 000 kg	10 000	8 000	20 000	14 000	10 000	7 000	5 000	14 000	11 200	10 000	8 000

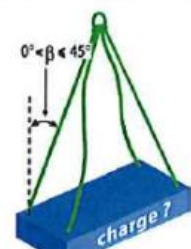
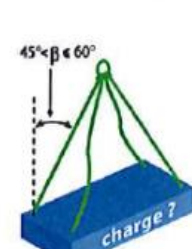
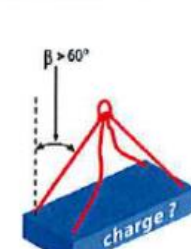
Elingage 2 brins : Chaînes de levage à maillons courts de classe 8 conformes à la norme EN 818-4 avec crochet type BK :



					
		Meilleur	Très bon	Bon	Interdit
Ø chaîne	CMU/Brin	Palonnier	$0^\circ < \beta \leq 45^\circ$	$45^\circ < \beta \leq 60^\circ$	$\beta > 60^\circ$
10	Chaîne 3 T 150	Maxi 6 T 300	Maxi 4 T 250	Maxi 3 T 150	
13	Chaîne 5 T 300	Maxi 10 T 600	Maxi 7 T 500	Maxi 5 T 300	
16	Chaîne 8 T 000	Maxi 16 T 000	Maxi 11 T 200	Maxi 8 T 000	
20	Chaîne 12 T 500	Maxi 25 T 000	Maxi 17 T 000	Maxi 12 T 500	

Elingage 4 brins (Identiques): Chaînes de levage à maillons courts de classe 8 avec crochet type BK :



				
		Très bon	Bon	Interdit
Ø chaîne	CMU/Brin	$0^\circ < \beta \leq 45^\circ$	$45^\circ < \beta \leq 60^\circ$	$\beta > 60^\circ$
10	Chaîne 3 T 150	Maxi 6 T 700	Maxi 4 T 750	
13	Chaîne 5 T 300	Maxi 11 T 200	Maxi 8 T 000	
16	Chaîne 8 T 000	Maxi 17 T 000	Maxi 11 T 800	
20	Chaîne 12 T 500	Maxi 26 T 500	Maxi 19 T 000	

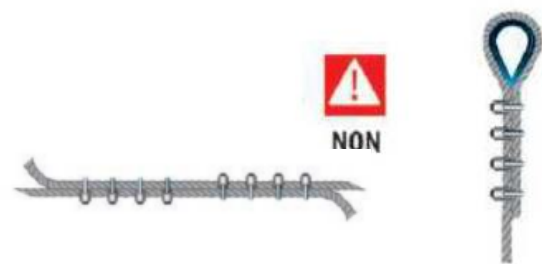
Conformément à la norme EN 818-4, 3 brins sont supposés reprendre la charge, charge rigide
Elingage 2 brins (identiques) : Câbles



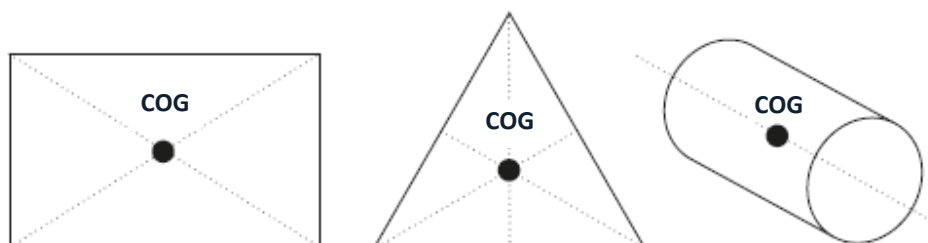
Ø câble	CMU/Brin	Palonnier	$0^\circ < \beta \leq 45^\circ$	$45^\circ < \beta \leq 60^\circ$	$\beta > 60^\circ$
22	Câble 5 T 500	Maxi 11 T 000	Maxi 7 T 700	Maxi 5 T 500	
29	Câble 9 T 500	Maxi 19 T 000	Maxi 13 T 300	Maxi 9 T 500	
34	Câble 14 T 000	Maxi 28 T 000	Maxi 19 T 500	Maxi 14 T 000	
38	Câble 17 T 500	Maxi 35 T 000	Maxi 24 T 500	Maxi 17 T 500	

3.3. Wire rope sling assembly

Use of wire rope clips in lifting assemblies that sustain loads is prohibited.



3.4. Evaluate the centre of gravity (COG)



It is preferable to sling above the centre of gravity for the stability of the load and special care should be made on that matter for loads with uneven shape/load distribution.

3.5. Attaching the load

- if the load is fitted with lifting points, the Slinger uses them in compliance with the technical characteristics and conditions for use of each lifting accessory (hook, shackle, threaded pin, etc.),

- the Slinger takes particular care to ensure safety latches work correctly,
- if the load is not fitted with attachment points, the Slinger uses the most appropriate accessories and slinging method (basket hitch or choker hitch ...) by applying the slinging mode factor to the WLL of accessories used,
- before slinging, the Slinger must know the methods and means to be employed to remove the slinging accessories safely

3.6. Areas of vigilance

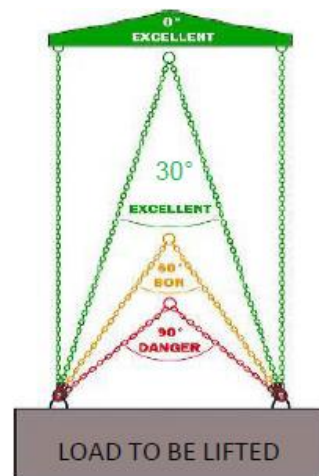
Slings angle

The load applied to slings varies according to the angle measured between the vertical and the sling.

The maximum angle of inclination allowed is 60°.

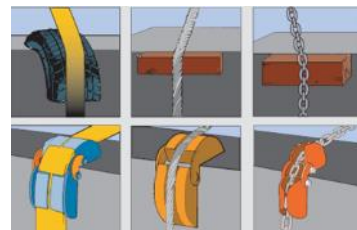
Position of hook safety latches

The safety latch of a hook is always facing upward.



Sharp edges

The load may have sharp edges. If this is the case, edge protection will be necessary to prevent damaging the slings.



Load balance and stability

First of all, apply tension to the sling without lifting the load. This step is to verify that the lifting assembly as a whole (crane hook, ring, slings, etc.) is properly positioned. Next, if everything is in place, lift the load slowly until it is clear from the ground/surface and confirm stability – if OK, the lifting operation can proceed normally

Eye bolts

Eye bolts are FORBIDDEN on precast concrete elements (such as segments, L walls, noise barriers etc.) and on wooden elements, even if they pass through the elements (bolt and nuts).

4. Management of lifting accessories

4.1. Lifting accessories register

A register for lifting accessories is set up and kept up to date by the project. Responsibilities for the maintenance of this register and, more generally, for the management of lifting equipment, are to be defined by the project in accordance with its organisation.

For each lifting accessory, the register should indicate the following minimum information:

- type of accessory
- identification number
- WLL
- date of purchase
- date of putting into service
- date of inspection
- date of expiry – for legally required inspections with certifications
- assignment to

On some sites, this register is managed by the lifting accessory supplier.

4.2. Marking

- lifting accessories are marked such that their essential characteristics for safe and proper use can be identified (as minimum ID/WLL and colour coding to indicate its relevant inspection period)
- permanent lifting accessory assembly is clearly marked to enable the user to identify its characteristics

4.3. Equipment and accessories examination

Lifting equipment and accessories are thoroughly examined:

- before being put into service, i.e. when used for the first time (new or second-hand equipment)
- before being put back into service, i.e. when the equipment has been dismantled, modified or one of its essential parts has undergone a major repair/transformation due to failure/breakdown

The purpose of these thorough examinations is to ensure that the equipment meets the prescribed specifications (which may be those set out in the manufacturer's instruction manual) and that it can be used safely.

The Project Manager/Project Director and/or the Plant Manager of the project or his representative on site ensures :

- that the declaration or certificate of conformity issued at the time of sale, hire, transfer or provision of the equipment is valid
- that the equipment meets the specifications of the purchase order

- that the equipment is suitable for the work to be carried out
- that the equipment and accessories are assembled and installed in accordance with manufacturer's specifications
- that any remarks in the last periodic general inspection report have been resolved or remedied
- that the equipment has an up-to-date maintenance log
- that measures are taken to prevent identified safety devices from being disabled or switched off
- that the equipment is operating efficiently as intended:
 - by conducting a static test of the lifting equipment, with all its accessories and its supports, to its maximum working load, multiplied by the static test coefficient, for a specified period
 - by conducting a dynamic test, which consists of having the lifting equipment move the maximum working load multiplied by the dynamic test coefficient, in such a way as to bring this load into all the positions it can occupy, without regard to the movement speed or the overheating of the lifting equipment

4.4. Periodic general inspection

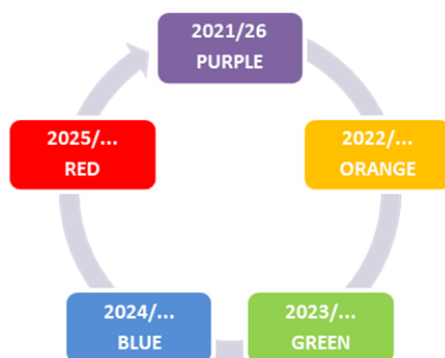
Lifting equipment and accessories are subject to periodic general inspections, the purpose of which is to detect damage or deterioration liable to create a hazard. These inspections are carried out by competent and independent persons.

The Project Manager/Project Director and/or the Plant Manager of the project or his representative on site sets the lifting equipment and accessories periodic general inspection schedule, considering that inspection are to be conducted:

- at intervals specified by the regulation in force
- at least every 12 months
- at least every 6 months for mobile lifting equipment (mobile cranes, forklifts, crane trucks)

The Lifting Designated Person coordinates the on-site periodic general inspections as per set schedule and, once an inspection is completed, ensures that a colored seal is affixed (paint, aluminium ring, flex ties, etc.) to said equipment/accessory in accordance with the project colour coding system. It is recommended, if possible, to refer to BYCN Material's cyclical yearly identification colour code (see below). A second sealing of the same color is required if 2 half-yearly inspections are carried out in the same year. The same process and approach is applied if more frequent inspections are legally required (for example quarterly).

This seal makes it easy to identify whether the equipment used on site at a given time has been inspected as "compliant" and is therefore authorized for use.



BYCN Material's cyclical yearly identification colour code

The Project Manager/Project Director and/or the Plant & Equipment Manager of the project or his representative on site, named in the *Lifting Management Plan*, keeps record of the periodic general inspection results, ensure they are available for audit (hard copy or digital) and that there is a copy in each lifting equipment at all times.

4.5. Visual inspection of equipment and accessories

Before starting any lifting operation, Signallers and Slingers carry out visual checks of lifting equipment and accessories that will be used for the operation and prevent use or immediately discard:

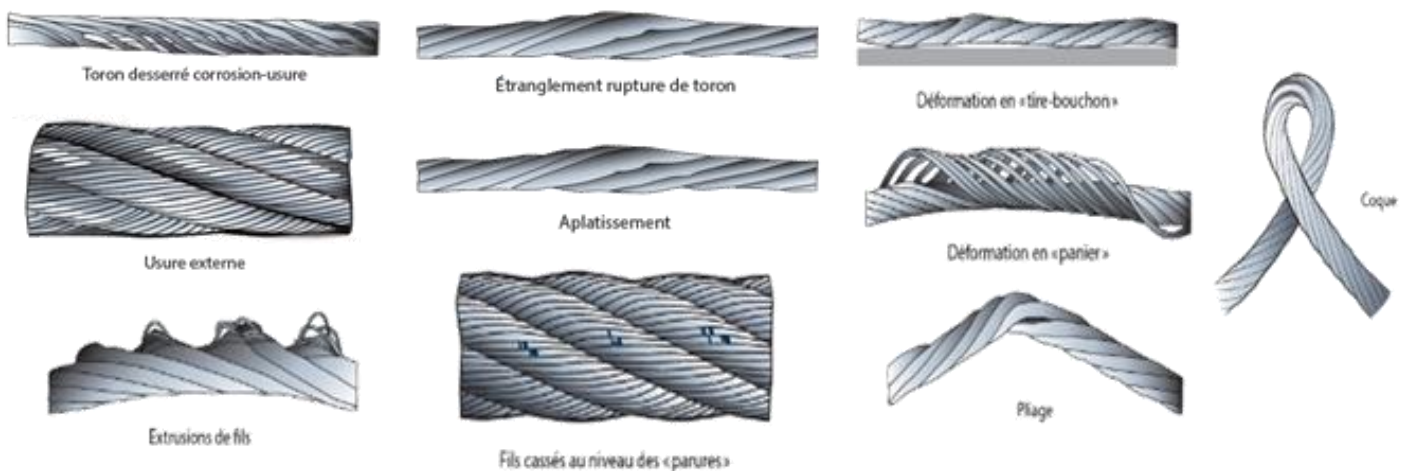
- any lifting equipment that does not have an up-to-date periodic general inspection
- any lifting accessory that does not have the current color identification
- any lifting accessory that does not have a WLL marking
- any lifting equipment and accessories that is damaged or degraded

The presence of a "compliant" seal does not exempt the Signallers and Slingers from having to carry out these visual checks.

4.6. Wire rope slings

They should be removed from service if the following are observed :

- wear, distortion or cracking of the upper or lower terminations
- a concentration of 3 broken outer wires in one strand
- a concentration of 6 broken outer wires over a length of 14 times its diameter
- a significant deformation of the wire rope with or without steel core protrusion
- a reduction in the wire rope diameter by up to 10% of its original diameter
- heat damage (discolouration)
- loss of lubrication or pitting of the wires caused by electric arcing



4.7. Chain slings

They should be removed from service if the following are observed :

- distortion of the upper or lower terminal fittings

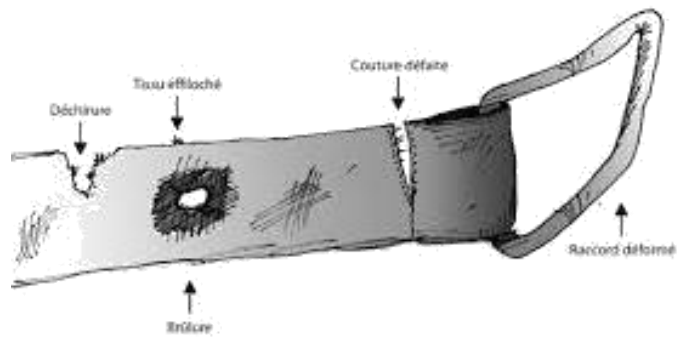
- DANGER**
-
- The diagrams illustrate various hazards associated with chain use. The top row shows a chain link being pulled apart, a chain link being twisted, a chain link being bent, and a chain link being crushed. The bottom row shows a chain link being pulled apart, a chain link being twisted, and a chain link being bent. Red arrows indicate the direction of force or movement. A red circle with a white exclamation mark is in the top left corner.



They should be removed from service if the following are observed :

-





Single-use webbing slings or slings attached to material/cargo delivered at site must be collected and discarded immediately.

4.9. Storage of lifting accessories

To protect lifting accessories from premature deterioration (corrosion, etc.), they must be stored under ideal conditions away from adverse weather.

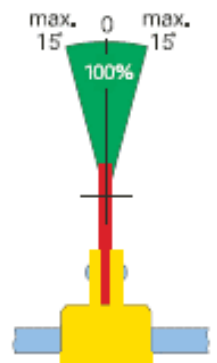
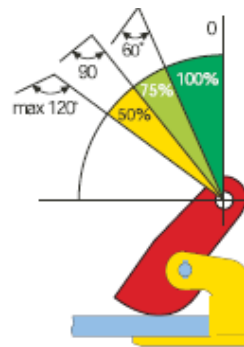
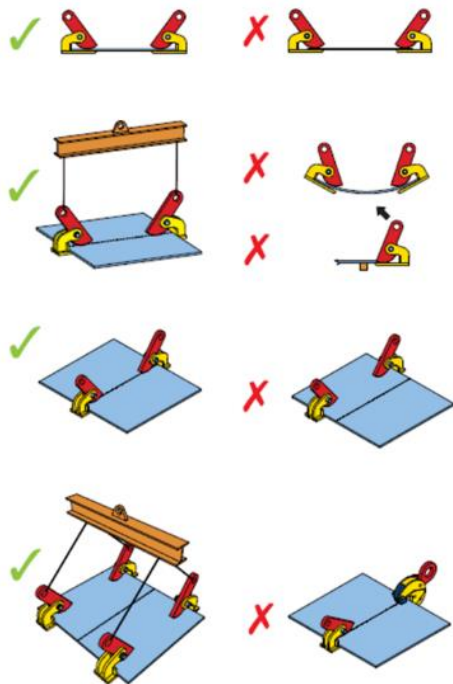
On site, the following measures must be put in place:

- either store the accessories in Site Supervisors' containers,
- or put them directly in "lifting containers" provided for such purpose



5. Specific lifting equipment and accessories

5.1. Metal sheet handling clamps



5.2. Vacuum spreader beam and vacuum pump

- Quick and accurate installation
- No time wasted on slinging
- No rolling platform required to unhook the load
- Remote control in the machinery cabin
- Ergonomics, no load handling
- Only one worker at the foot of the machinery

Gain in productivity :



- 100 panels/day with slings
- 250 panels/day with lifting beam



5.3. Remote controlled hooks

Reducing the risks of accidents, their severity and musculoskeletal disorders.



5.4. Lifting magnets and electromagnets



5.5. Segment lifters, barrel lifters, jib cranes

Inspection instructions are similar to those mentioned above for accessories.



5.6. Eye bolt

Eye bolts shall always be considered as the last option by Design/Methods/Operations.



Eye bolts are FORBIDDEN on precast concrete elements (such as segments, L walls, noise barriers etc.) and on wooden elements, even if they pass through the elements (bolt and nuts).

A recommended alternative is the use of Arteon lifting hands, which allow for the safe lifting of precast concrete elements (such as slabs, panels, blocks, etc.) by means of dedicated anchors inserted in advance.



Eye bolt in lifting systems could be considered if :

- There is no other technical alternatives
- They are part of the equipment (for instance, an engine supplied with its own eye bolts)
- When they pass through an equipment, they are reachable, bolt can be correctly tightened and remain captive
- The type and specifications of the eye bolts are specifically defined
- Specific lifting instructions and requirements are issued
- Users and supervision are trained
- Inspection and maintenance plans are in place

5.7. Articulated hoist rings

This type of anchorage is designed for temporary or permanent applications and can be used for lifting (winch, hoist, pulley hoist, etc.). It can be used with manual or motor hoists.

Advantages

- lifting in variable directions : the ring can rotate and swivel
- anchorage of hoist point can be designed with the manufacturers' software
- pre-assembled element (one piece), no need to assemble the components

Checks and vigilance before use

- the anchor point must be identified in color and with a sticker
- the maximum load must be displayed near the anchor point with its WLL



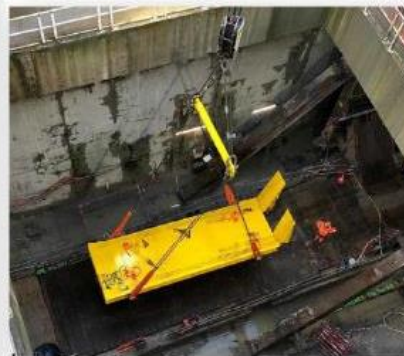
Inspections

This type of anchorage is a lifting accessory that requires periodic general inspections and non-destructive tests shall be carried out each time after the system is installed, adjusted or modified to guarantee its capacities before use :

- by an external certification body
- or in-house, using a dynamometric test device supplied by the manufacturer. The test is carried-out by a trained and competent person.



5.8. Construction plants lifting platforms



5.9. Big bags

Lifting of big bags must be carried using lifting baskets.



Max filling height = 2/3



Use self-emptying buckets when possible



Do not re-use "single-use" bags



Do not use big bags to handle rebars or scrap metal

5.10. Storage baskets

- Baskets are closed on 5 sides and adapted to the load to be lifted
- Mandatory strapping for baskets non closed on 5 sides
- The contents of baskets shall not exceed the height of the sides
- The load must be centered in the basket and strapped if space exist between the load and the basket sides

Lifting

- Suitability inspection
- Condition of the basket
- All lifting points are used
- WWL is respected



Basket filling

- Material packed in the green section must not exceed the SWL of the basket
- When stacking baskets, ensure the stability of the assembly

